

Chapter 8: Active Inference in Continuous Time

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Introduction: Two Types of Time

- Discrete Time (Chapter 7)
 - Structure: Steps (t , $t+1$, $t+2$...)
 - Actions: Decisions / Choices (e.g., Left vs. Right)
 - Focus: Planning and Outcome selection.
- Continuous Time (Chapter 8) Structure:
 - Flow (Smooth dynamics)
 - Actions: Control / Movement (e.g., Muscle force)
 - Focus: Tracking dynamics and Real-time execution.

The Challenge of Real-World Physics-

- The Nature of Reality
 - The physical world does not wait for us to make decisions.
 - Objects move, gravity acts, and light fluctuates continuously. '
- The Brain's Task
 - To survive, the brain must model these Dynamics.
 - It needs a way to represent Calculus (Rates of change) in neural circuits.

Defining the Generative Process (The World)-

- Stochastic Differential Equations
 - We model the world using equations of motion.
 - Equation: $\dot{x} = f(x, u) + \omega$
- Components
 - x (Hidden States): True physical variables (e.g., Position, Temperature)
 - u (Action): Forces applied by the agent.
 - w (Noise): Random fluctuations in the world
 - f (Flow Function): The "Physics" describing how states change

Defining the Generative Model (The Brain)

- The Brain's Simulation
 - The brain maintains its own version of the physics equations
 - Equation: $\dot{\tilde{x}} = f(\tilde{x}) + \tilde{\omega}$
- Key Difference
 - The brain doesn't have direct access to x (World State)
 - It only has access to y (Sensory Input).
 - Goal: Update internal belief to match the hidden cause x using only y

Solution: Generalized Coordinates of Motion (1)

- The Concept
 - How do we represent a moving object at a single instant?
 - Just knowing "Position" is not enough to predict the future.
- The Vector Representation
 - The brain encodes a "snapshot" of the entire trajectory.
 - State vector $\tilde{x} = [\text{Position, Velocity, Acceleration, Jerk...}]^T$
 - This allows the brain to hold the "motion" itself in a static moment.

[Slide 7] Solution: Generalized Coordinates of Motion (2)

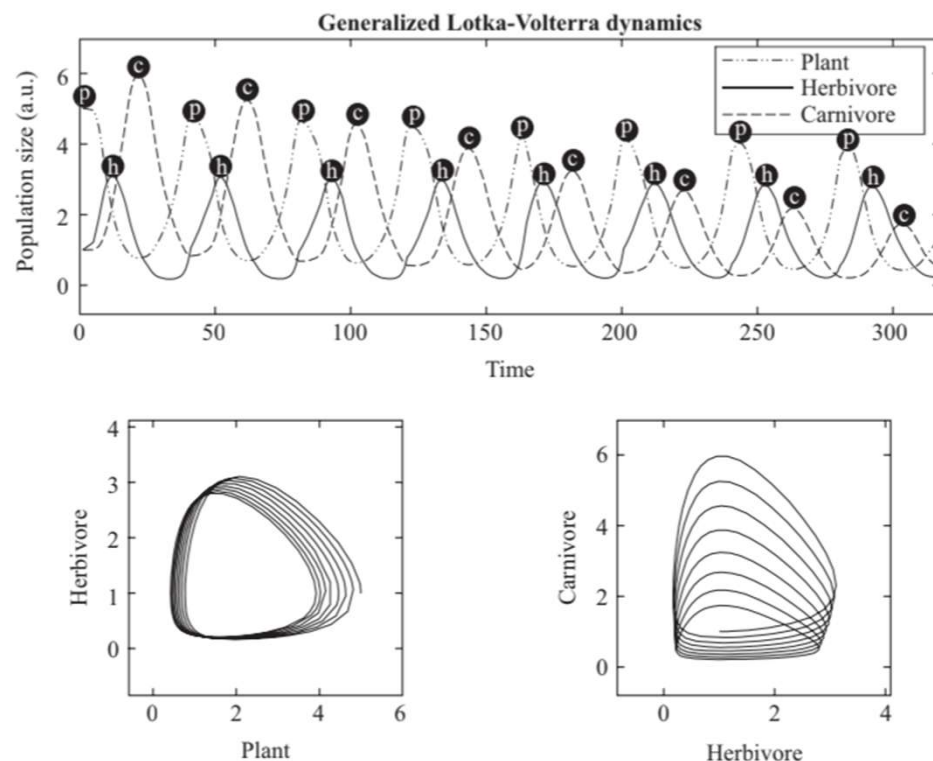
- **Why Infinite Derivatives?**

- By keeping track of higher-order derivatives (Velocity, Acceleration...), the brain can **predict the immediate future** without waiting.

- **Taylor Series Expansion**

- Allows "**Instantaneous**" **prediction** of where the object will be next.
- Encodes the **Flow of Time** into a static neural code.

[Slide 8] Perception: Predictive Coding



- The Mechanism
- **Prediction Unit (Blue):** Sends predictions ($g(\mu)$) **DOWN**.
- **Error Unit (Red):** Sends prediction errors (ε) **UP**.
- The system settles when errors are minimized.

(Parr et al., 2022, Figure 8.2: The Predictive Coding Scheme)

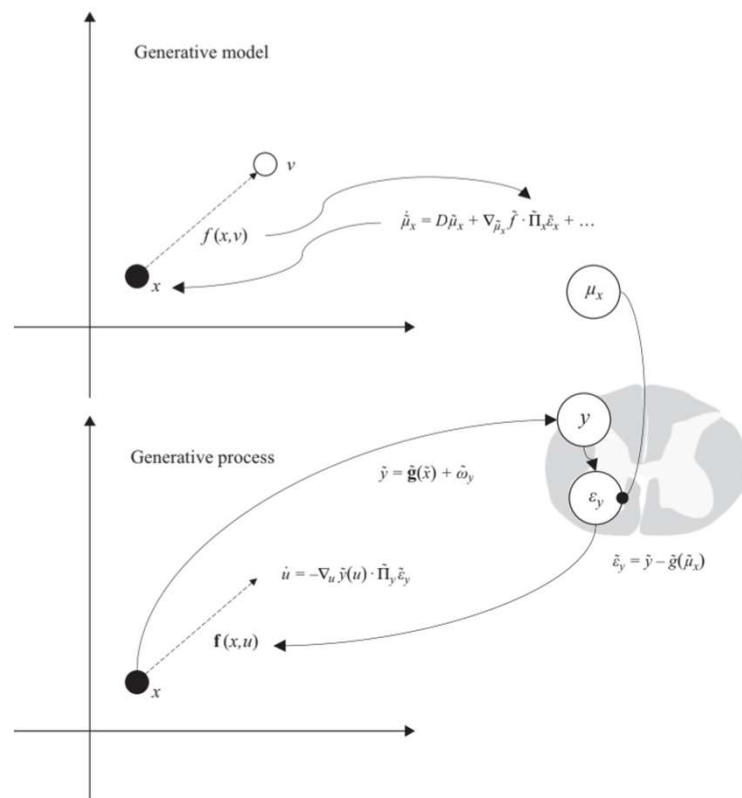
Perception as Gradient Descent

- **Minimizing Free Energy (F)**
 - Mathematically, F is a function of **Prediction Error**.
 - To minimize F , the brain changes its **Internal Belief (μ)**.
- **Gradient Descent Dynamics**
 - $\dot{\mu} = D\mu - \frac{\partial F}{\partial \mu}$
 - **Meaning:** "Update the belief in the direction that **reduces the error**."
 - Perception is simply **sliding down** the error landscape.

Action in Continuous Time

- **Comparison**
 - **Discrete:** "Select" a Policy (**Plan**)
 - **Continuous:** "Execute" a Reflex (**Action**)
- **Action as Inference**
 - Action is just another way to minimize Prediction Error.
 - Instead of updating the **Brain** (Perception)...
 - We update the **World** (Action).

The Spinal Reflex Arc



(Parr et al., 2022, Figure 8.1)

• The Mechanism

- **Top-down:** Brain sets the **Target Point** (Prediction).
- **Bottom-up:** Spinal cord detects **Error** (Difference).
- **Reflex:** Motor neurons fire to **Contract Muscle**.
- **Outcome:** Arm moves to Target → **Error becomes Zero**.

Action vs. Command

- **How Action Works**

- The brain does **NOT** send motor commands
(e.g., "Power 50%")
- The brain sends **Prediction**
(e.g., "My arm is at Position B")

- **The Conflict**

- **Reality:** Arm is at A.
- **Prediction:** Arm is at B.
- **Result:** A huge **Proprioceptive Prediction Error**.
- → Action resolves this conflict by moving the arm.

Sensory Attenuation

- **The Paradox of Movement**

- If the brain perfectly trusts the current sensory input ("Arm is at A")...
- It will simply update its belief to "Arm is at A".
- **Result: No Movement** (Prediction matches Reality).

- **The Solution: Attenuation**

- To move, the brain must **IGNORE** the current sensory data temporarily.
- It lowers the **Precision (Volume)** of the sensory error.
- **Motto:** *"Don't listen to the body; force the body to obey the prediction."*

Hierarchical Dynamic Models

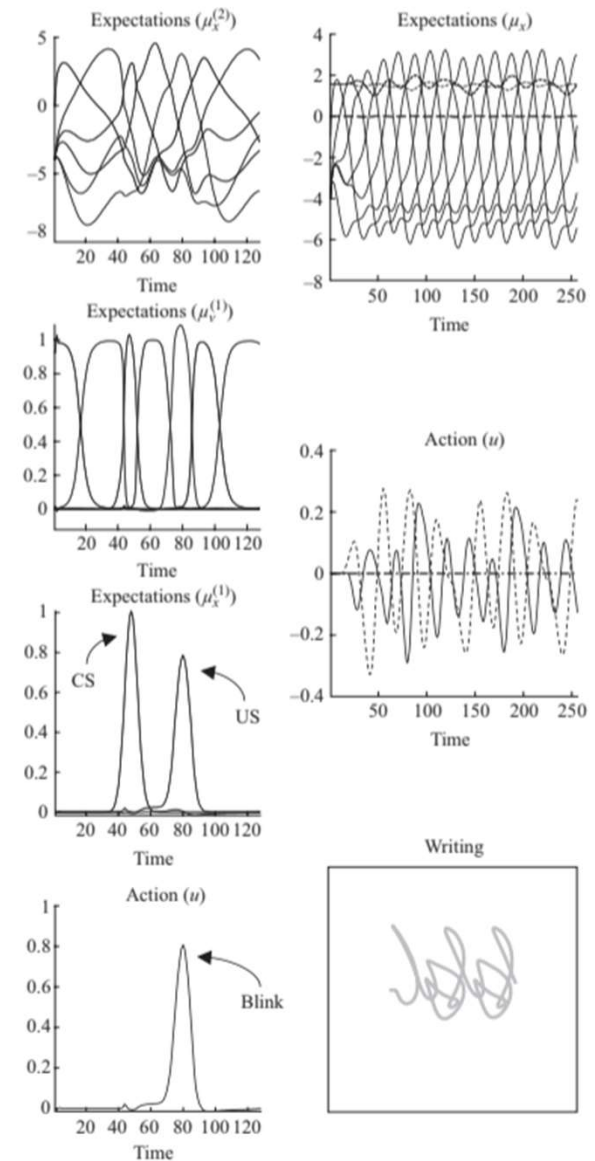
- **Stacking the Dynamics** The brain is not just one layer of differential equations.
- It is a **Hierarchy** of dynamical systems.
- **Levels of Abstraction**
 - **Higher Levels (Slow):** Intentions / Narratives
 - **Lower Levels (Fast):** Reflexes / Saccades / Physics

Top-Down vs. Bottom-Up

- **Interaction**

- **Top-Down:** Provide the "Trajectory" for the level below.
 - (e.g., "Wave Hello")
- **Bottom-Up:** Pass up the "Prediction Error" unresolved by physics.
 - (e.g., "Arm is heavy")

(Parr et al., 2022, Figure 8.3)



Conclusion

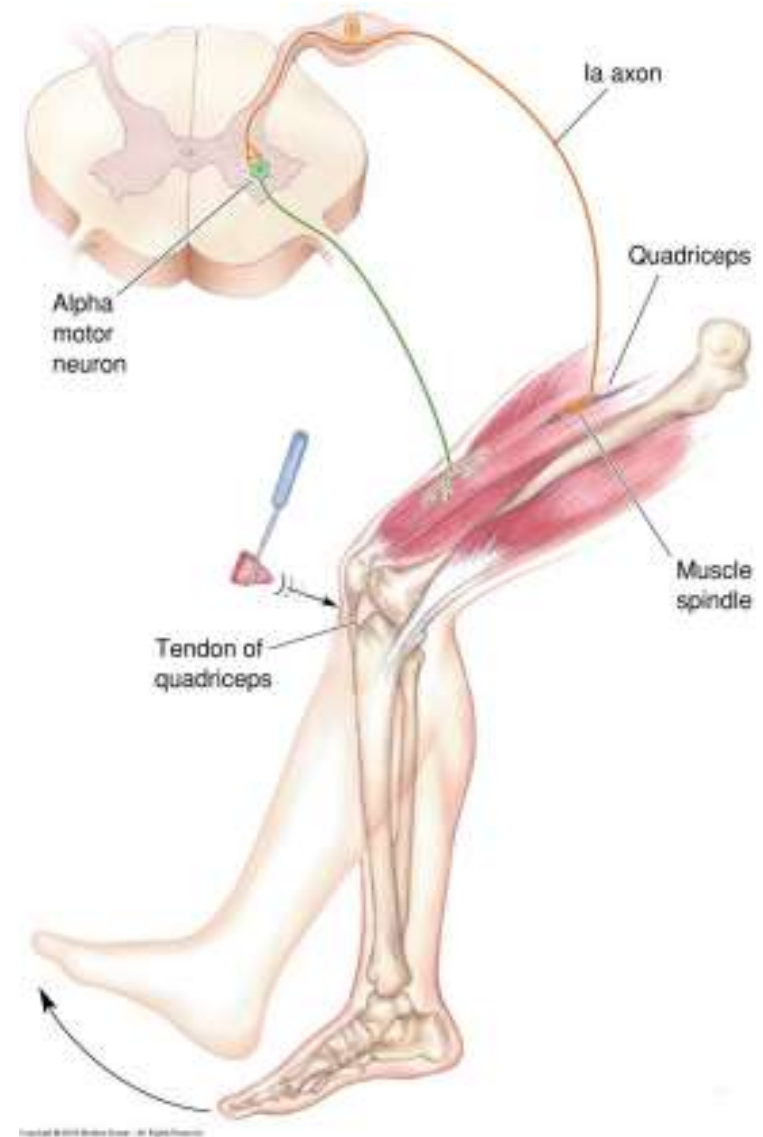
- **Summary**

- **Physics:** Modeled via **Generalized Coordinates** (Trajectories).
- **Perception: Predictive Coding** (Tracking the world).
- **Action: Reflex Arcs** (Forcing the world to match predictions).

- **Final Thought**

- The brain doesn't just process information; it **physically couples** with the world through continuous loops of Action and Perception.

Recursive Arc



Discussion 1: The Thickness of the Present

Topic: Generalized Coordinates & Phenomenology

- Context:
 - The brain encodes position (x), velocity (x'), and acceleration (x'') simultaneously.
- **Implication:**
 - To know the velocity is to know the **immediate future**.
 - Therefore, our perceptual "Now" is not a 0-second point.
 - It has a "**Temporal Thickness**" that bridges the past and future.
 - *Do we perceive the future directly?*

Discussion 2: The Paradox of Movement

- **Topic: Healthy Hallucinations?**
- **Context:** To move, we must attenuate (ignore) sensory evidence that says "I am still."
- **Question:**
 - Is voluntary movement a form of "**Self-fulfilling Prophecy**" or "**Controlled Hallucination**"?
- **Pathology:**
 - **Too much reality?** → Parkinson's (Frozen)
 - **Too much prediction?** → Schizophrenia (Hallucination)

Discussion 3: Reward vs. Belief

Topic: What drives us?

- **Context:**

- **Standard View:** We maximize **Reward** (Cost Function).
- **Active Inference:** We minimize **Surprise** (Prediction Error).

- **Question:**

- Did you eat lunch to get **Pleasure**?
- Or did you eat to fulfill your **Prior Belief** that *"I am a living agent who eats"*?
- Does this change how we understand **Motivation**?